

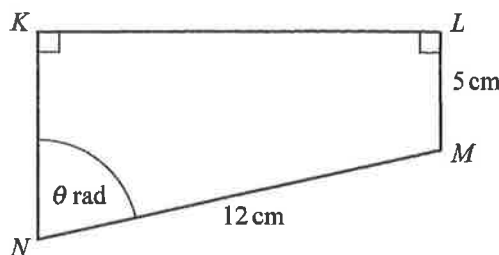
Basic

- Given that $y = Ae^{2x} + Be^{-x}$, and that $\frac{dy}{dx} + 4y = e^{2x} - e^{-x}$, find the value of each of the constants A and B . [4]
[N19/I/3]
- A particle moves along the curve $y = \ln(3x - 1)$ in such a way that the y -coordinate of the particle is increasing at a constant rate of 0.06 units per second. Find the rate at which the x -coordinate of the particle is increasing at the instant when $x = 7$. [3]
[N17/II/8(a)]

Intermediate

- The curve $y = 5 - e^{2x}$ intersects the x -axis at A and the y -axis at B .
 - Find the coordinates of A and B , in logarithmic form where appropriate. [2]
 - Find the gradient of the curve at A . [2]
 [N22/II/6(b)]

4.



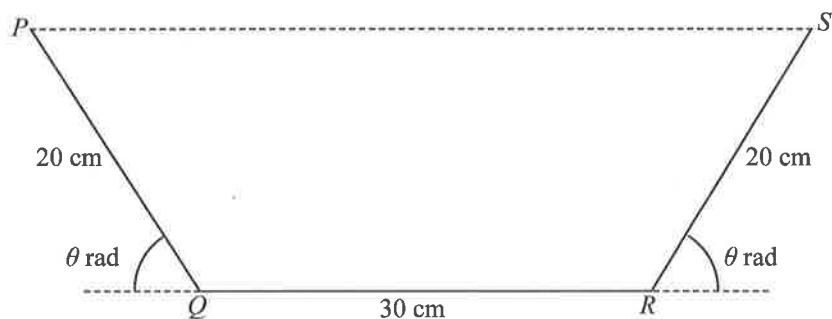
The diagram shows a metal plate in the shape of a trapezium in which angles NKL and KLM are right angles. The lengths of LM and MN are 5 cm and 12 cm respectively. The acute angle KNM is θ radians.

- Show that the perimeter, P cm, is given by $P = 22 + 12 \cos \theta + 12 \sin \theta$. [2]
 - Find the value of R when $12 \cos \theta + 12 \sin \theta$ is expressed as $R \cos(\theta - \alpha)$, where R and α are constants, and hence state the maximum perimeter of the plate. [3]
 - Show that the area of the plate, A cm², is given by $A = 60 \sin \theta + 36 \sin 2\theta$. [2]
 - The maximum area of the plate is obtained when the value of θ , which can vary, gives a stationary value of A . Find this value of θ . [5]
- [N19/II/11]
- Find the value of the constant k for which $y = e^{-x}x^2$ is a solution of the equation

$$e^x \left(\frac{d^2y}{dx^2} + 2 \frac{dy}{dx} + y \right) = k. \quad [7]$$

[N17/II/1]

6.



The diagram shows the vertical cross section $PQRS$ of an open trough made from plastic sheeting. The lengths of PQ , QR and RS are 20 cm, 30 cm and 20 cm respectively. The trough rests with QR on horizontal ground and both PQ and RS are inclined at an angle θ radians to the ground.

- (i) Show that the area, $A \text{ cm}^2$, of the cross section $PQRS$ is given by

$$A = 600 \sin \theta + 200 \sin 2\theta. \quad [4]$$

- (ii) Given that θ can vary, find the value of θ for which the trough can hold a maximum amount of water. [5]

[N15/I/11]